


# SDEV2501

## Day 13 – Intro to Data Visualization

A LEADING POLYTECHNIC COMMITTED TO YOUR SUCCESS

# Today’s objective

- Define data visualization
- Differentiate between types of visualizations

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# Agenda

- Review steps of data wrangling
- What is data visualization?
- How to choose the right type of visualization

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# REVIEW

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# Review

- What are key tasks in data cleaning?
- Why does it matter?
- What happens if we don’t clean data?

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# DATA VISUALIZATION

“If you can dream it, think about it, visualize it, then you can manifest it.”

— Debasish Mridha

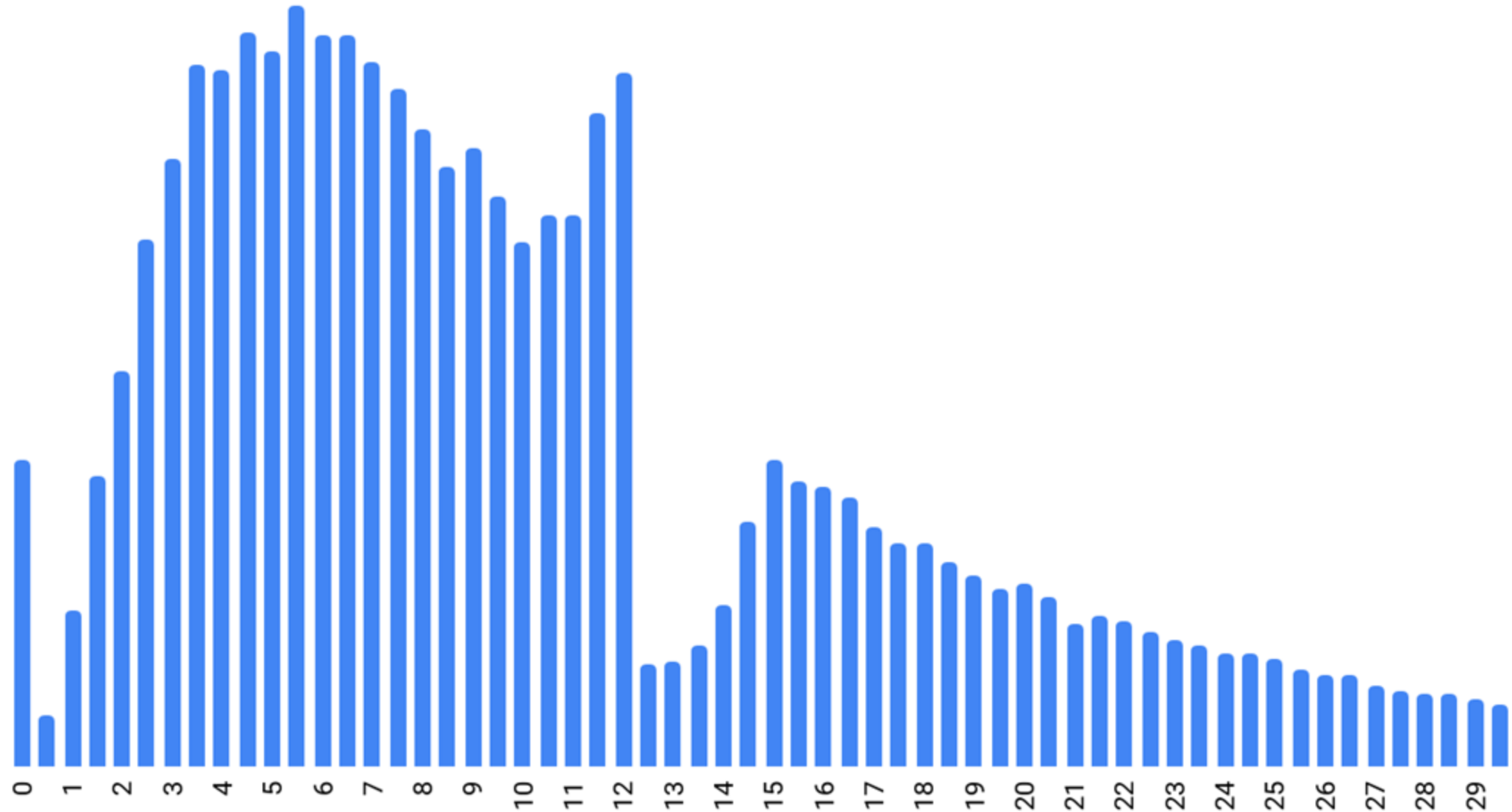
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# Why do we visualize data?

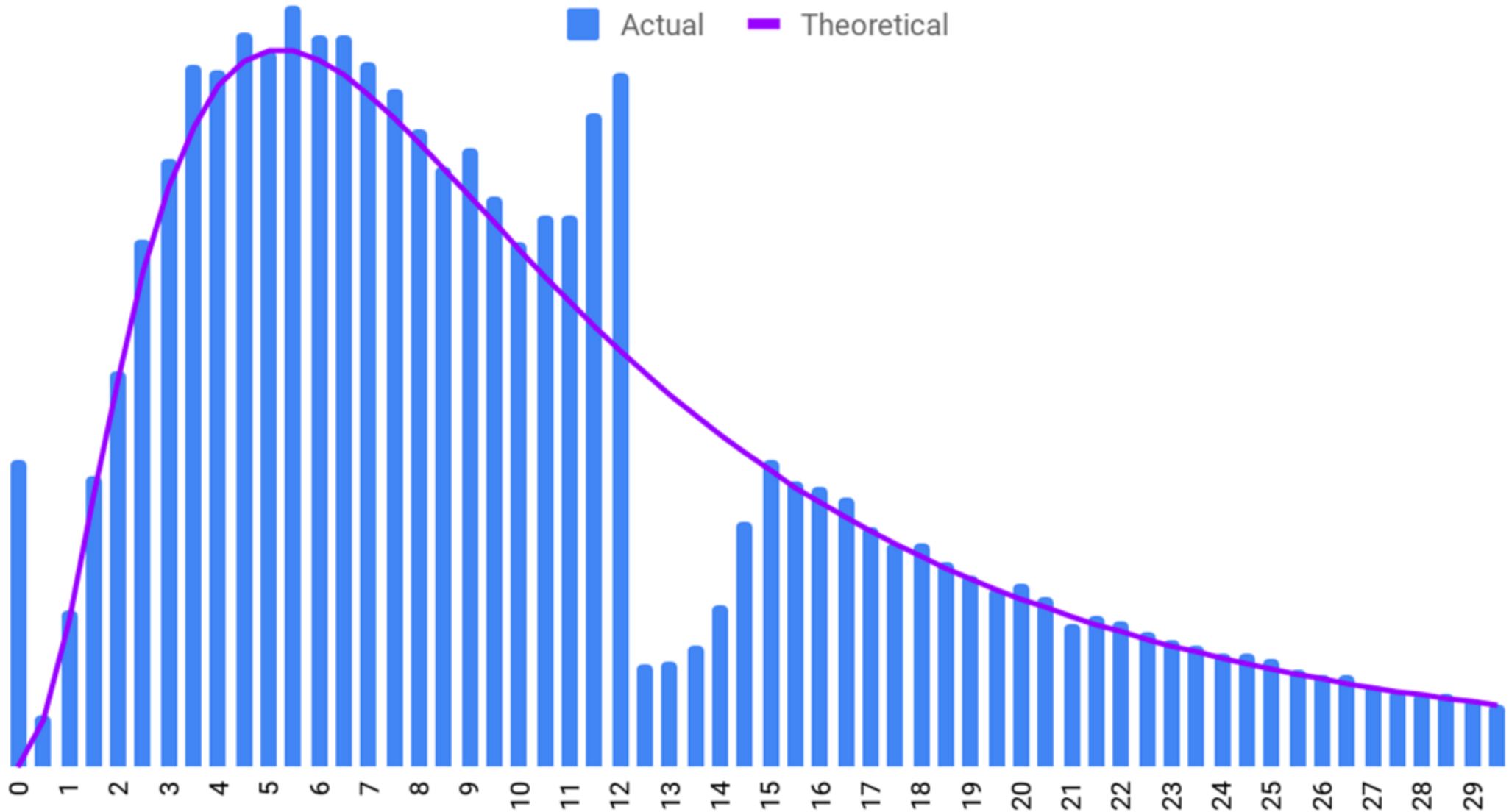
- Data visualization helps us **understand** and **explain** our data.

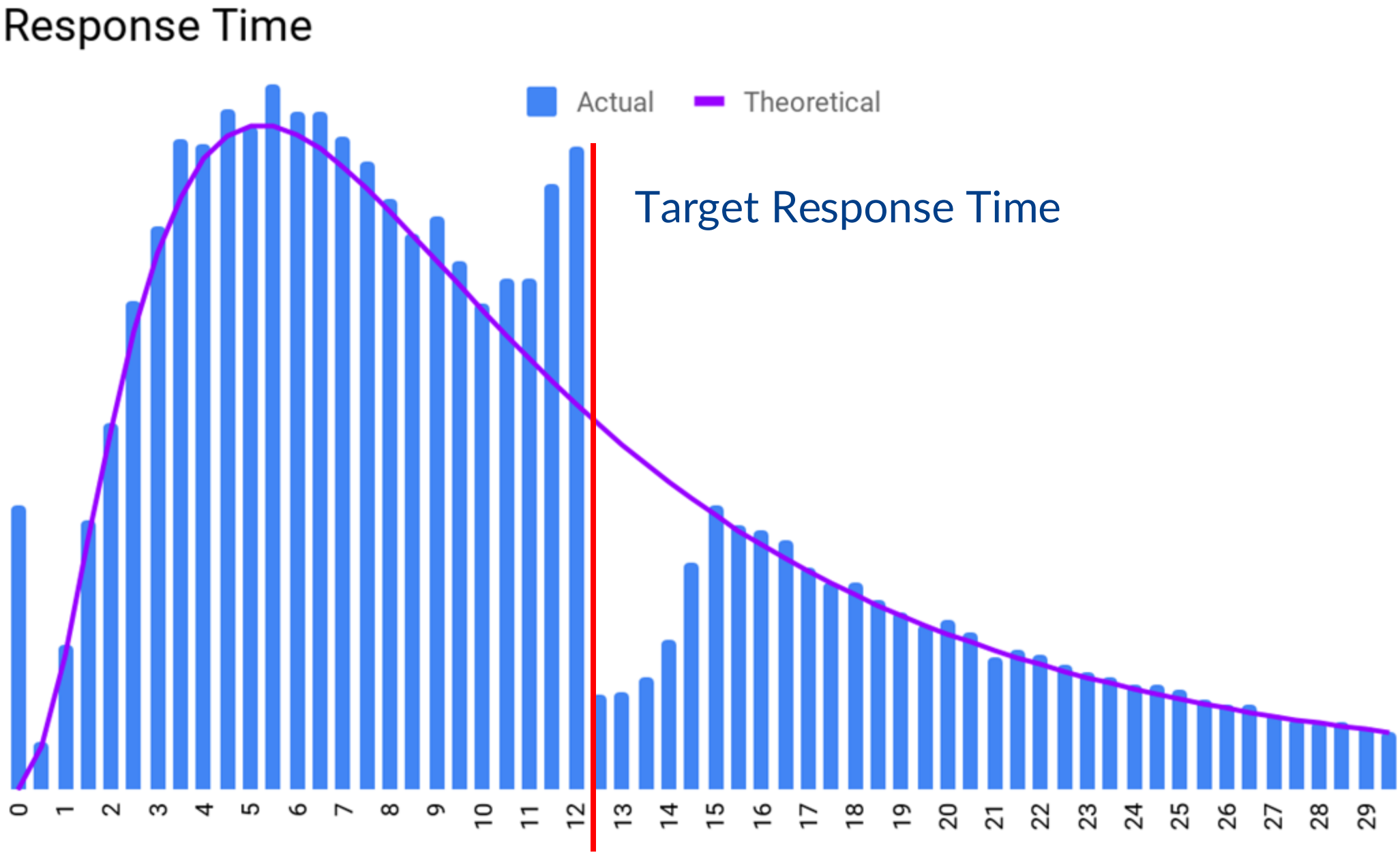
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# Response Time



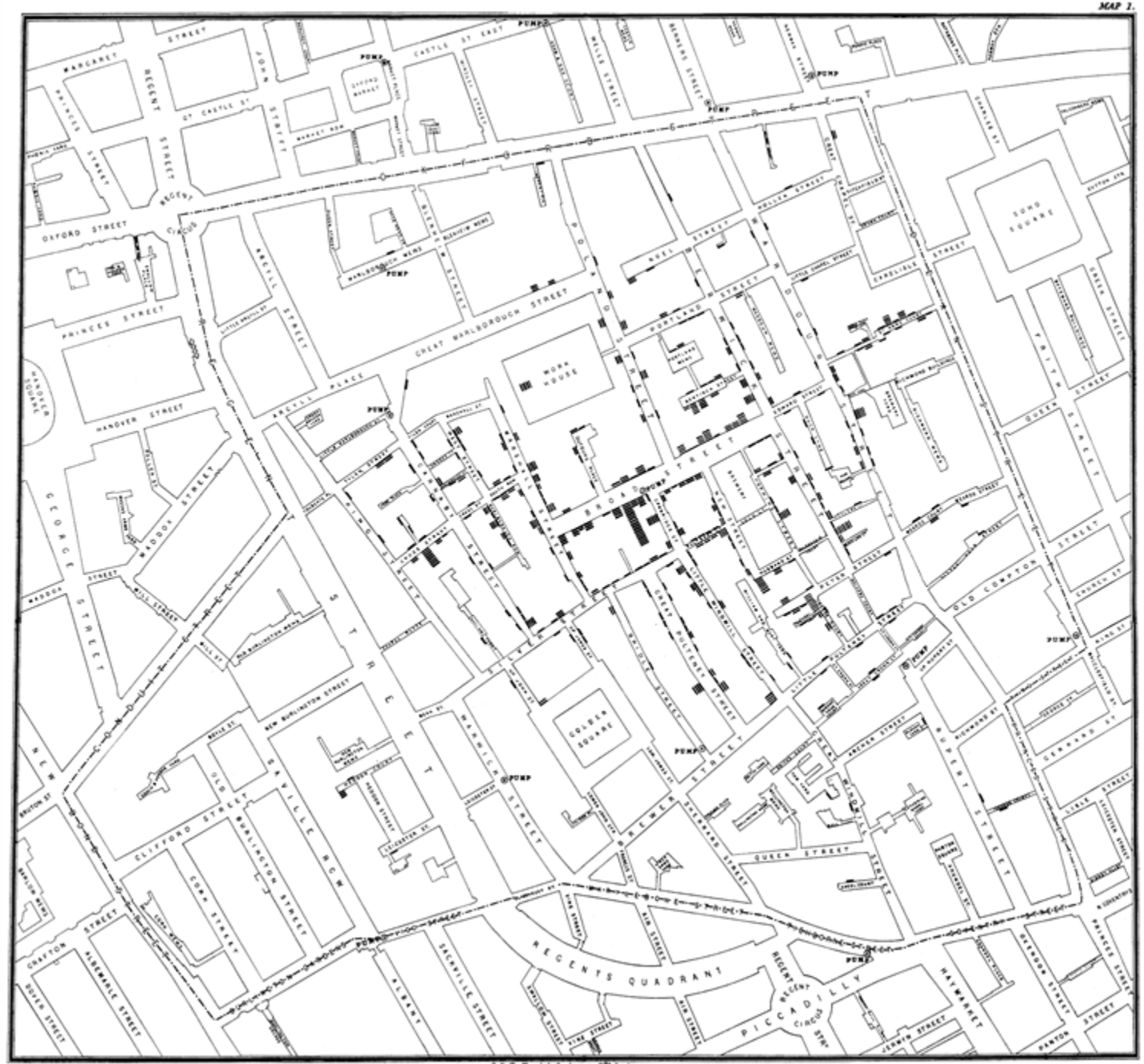
# Response Time




Response Time
Actual
Theoretical
Target Response Time
0
1
2
3
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10
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12
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26
27
28
29

# DATA VIZ CAN ALSO SAVE LIVES.

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C. F. Cheffins, Lith. Southampton St. London.

SCALE 30 INCHES TO A MILE.

# Common Visualizations

- Pie charts
- Line charts
- Scatter plots
- Bar charts

# How do we choose the right viz?



Depends on the question



Depends on the data type



data-to-viz.com

# SMALL GROUP CHALLENGE: HOW MANY WAYS CAN YOU COMMUNICATE 2 NUMBERS?

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75, 37

# Don’t proceed yet.

1

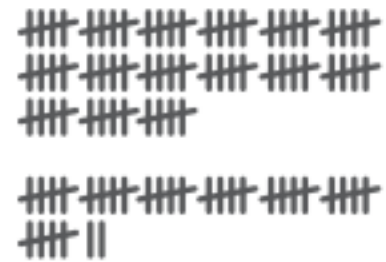
*75, 37*

*a*



*b*





*c*

2



3



4



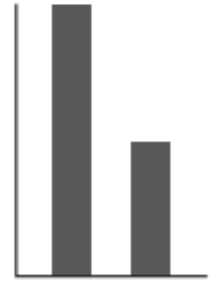
5



a

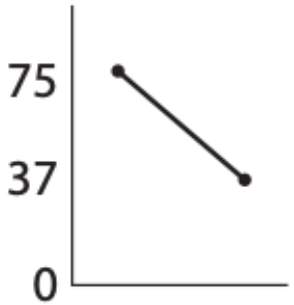


b



c

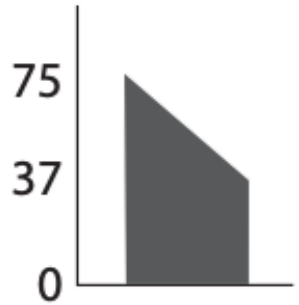
6




a




b

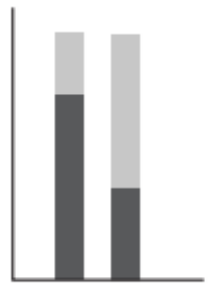



c

7

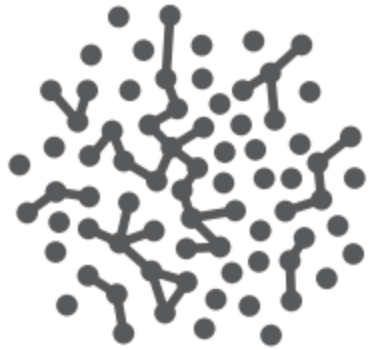


a



b

39



20

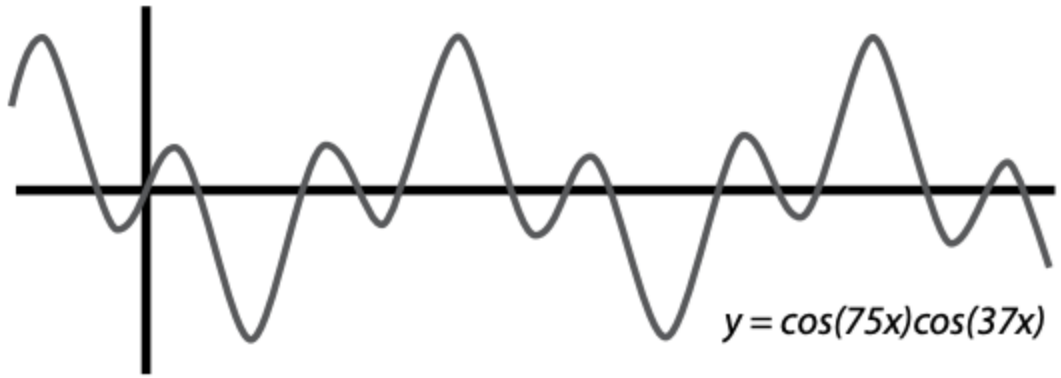


*a*

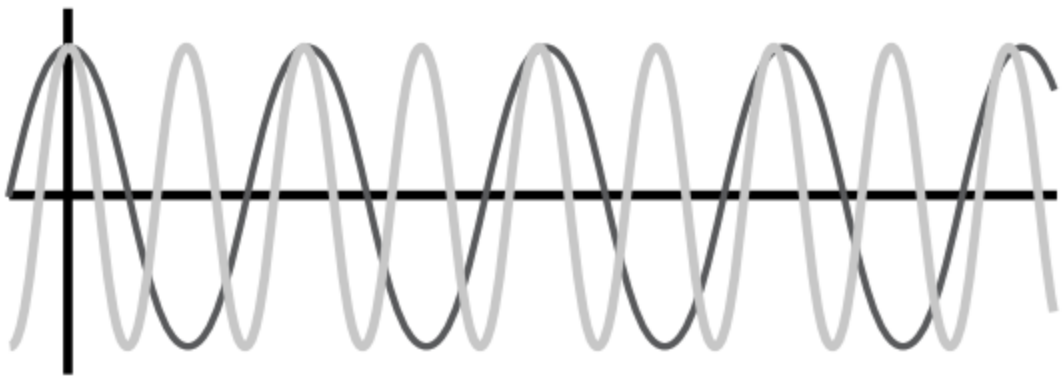


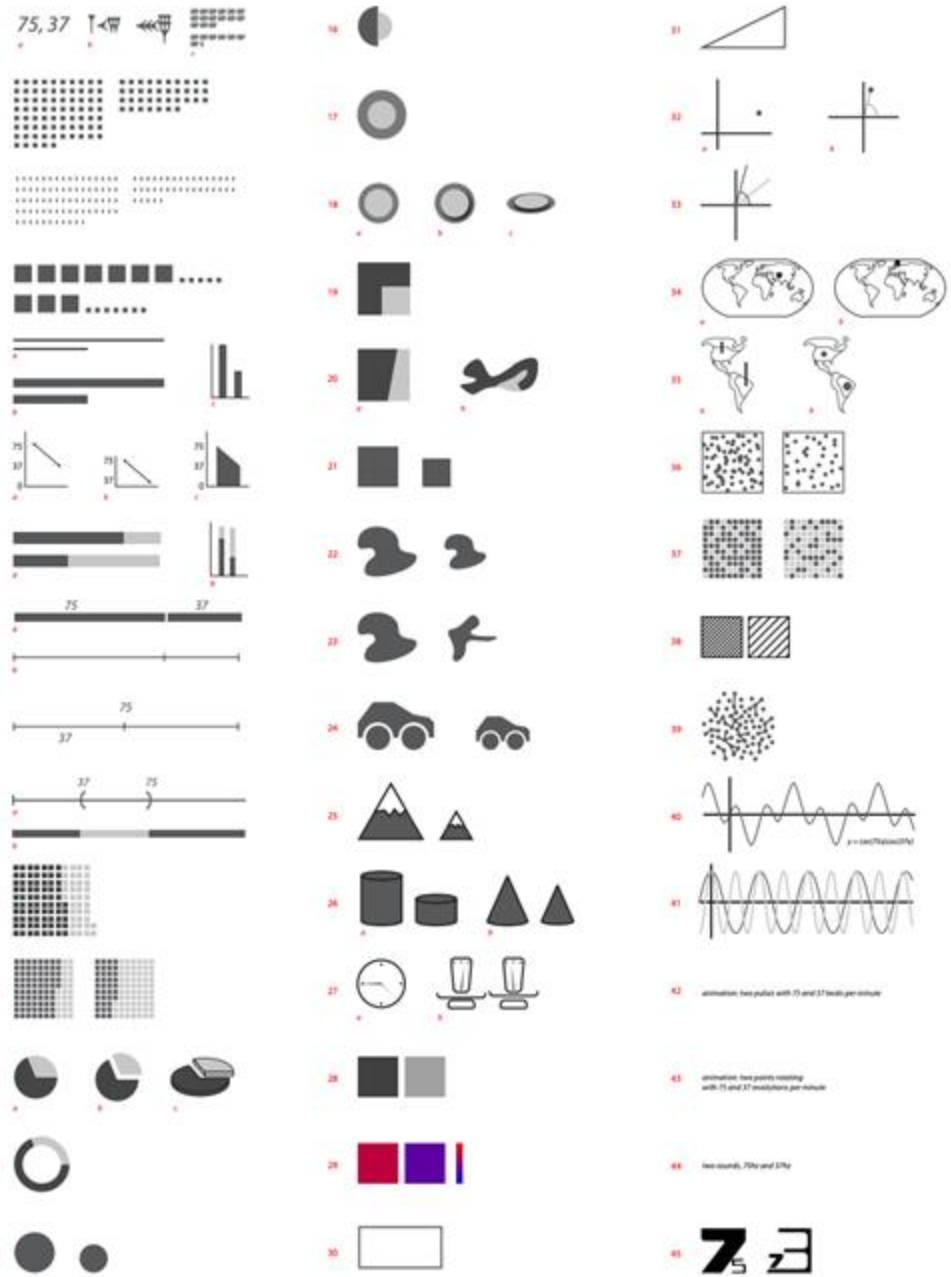
*b*

40




41




75, 37
75
37
y = sin(75πsin(37πt)
animation: two pulses with 75 and 37 beats per minute
animation: two points rotating
with 75 and 37 revolutions per minute
two sounds, 75hz and 37hz

# PRINCIPLE #1:
# SELECT THE RIGHT VISUAL FOR YOUR DATA, YOUR MESSAGE AND YOUR AUDIENCE

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# Visual vocabulary

*Designing with data*

There are so many ways to visualise data - how do we know which one to pick? Use the categories across the top to decide which data relationship is most important in your story, then look at the different types of chart within the category to form some initial ideas about what might work best. This list is not meant to be exhaustive, nor a wizard, but is a useful starting point for making informative and meaningful data visualisations.

ft.com/vocabulary

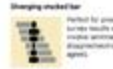
## Deviation

Emphasise variations (+/-) from a fixed reference point. Typically the reference point is zero but it can also be a target or a long-term average. Can also be used to show sentiment (positive/neutral/negative).

**Example FT uses**
Trade surplus/deficit, climate change

**Diverging bar**

A simple standard bar chart that can handle both negative and positive magnitude values.

**Diverging stacked bar**

Perfect for presenting survey results which involve sentiment (eg disagree/neutral/agree).

**Spine**

Splits a single value into two contrasting components (eg male/female).

**Surplus/deficit filled line**

The shaded area of these charts almost a balance to be shown - either against a baseline or between two series.

## Correlation

Show the relationship between two or more variables. Be mindful that, unless you tell them otherwise, many readers will assume the relationships you show them to be causal (i.e. one causes the other).

**Example FT uses**
Inflation and unemployment, income and life expectancy

**Scatterplot**

The standard way to show the relationship between two continuous variables, each of which has its own axis.

**Column + line timeline**

A good way of showing the relationship between an amount (column) and a rate (line).

**Connected scatterplot**

Usually used to show how the relationship between 2 variables has changed over time.

**Bubble**

Like a scatterplot, but adds additional detail by sizing the circles according to a third variable.

**XY heatmap**

A good way of showing the patterns between 2 categories of data. Also effective at showing fine differences in amounts.

## Ranking

Use where an item's position in an ordered list is more important than its absolute or relative value. Don't be afraid to highlight the points of interest.

**Example FT uses**
Wealth, deprivation, league tables, constituency election results

**Ordered bar**

Standard bar charts display the ranks of values much more easily when sorted into order.

**Ordered column**

See above.

**Ordered proportional symbol**

Use when there are big variations between values and/or seeing fine differences (whereas dot is not so important).

**Dot strip plot**

Dots placed in order on a strip are a space-efficient method of ranking set ranks across multiple categories.

**Slope**

Perfect for showing how ranks have changed over time or uses between categories.

**Lollipop**

Collapse (see more attention to the data value than standard bar/column and can also show more total value effectively).

**Rump**

Effective for showing changing rankings across multiple dates. For large datasets, consider grouping lines using colour.

## Distribution

Show values in a dataset and how often they occur. The shape (or 'skew') of a distribution can be a memorable way of highlighting the lack of uniformity or equality in the data.

**Example FT uses**
Income distribution, population (age/sex) distribution, assessing inequality

**Histogram**

The standard way to show a statistical distribution - keep the gaps between columns small to highlight the 'shape' of the data.

**Dot plot**

A simple way of showing the change or range (min/max) of data across discrete categories.

**Dot strip plot**

Good for showing individual values in a distribution, can be a problem when too many dots have the same value.

**Barcode plot**

Like dot strip plots, good for displaying all the data in a table. Also work best when highlighting individual values.

**Boxplot**

Summarise multiple distributions by showing the median (centre) and range of the data.

**Violin plot**

Similar to a box plot but can offer fans with complex distributions (data that cannot be summarised with values/ average).

**Population pyramid**

A standard way for showing the age and sex breakdown of a population (contraction effectively back to back histograms).

**Cumulative curve**

A good way of showing how wrapped a distribution is (y axis is always cumulative magnitude) - a kink is always a noticeable.

**Frequency polygons**

For displaying multiple distributions of data, like a regular line chart, best limited to a maximum of 3 or 4 datasets.

**Ternary**

Use to emphasise individual points in a distribution. Points can be sized to an additional variable. Best with medium sized datasets.

## Change over Time

Give emphasis to changing trends. These can be short (intra-day) movements or extended series (covering decades or centuries). Choosing the correct time period is important to provide suitable context for the reader.

**Example FT uses**
Share price movements, economic data series, sector changes in a market

**Line**

The standard way to show a changing time series. If data are irregular, consider markers to represent data points.

**Column**

Column's work well for showing change over time (or similar) with one series of data at a time.

**Column + line timeline**

A good way of showing the relationship over time between an amount (column) and a rate (line).

**Slope**

Good for showing changing data as long as the data can be simplified to two (or 3 points without missing a key part of story.

**Area chart**

Use with care - these are good at showing changes to total, but making changes to components can be very difficult.

**Candlestick**

Usually focused on day to day activity. These charts show open/close/high and low/close points of each day.

**Fan chart (projections)**

Use to show the uncertainty in future projections - usually two groups that fan from forward to projection.

**Connected scatterplot**

A good way of showing changing data for two variables whenever there is a relationship (see 'some in' of magnitude).

**Calendar heatmap**

A great way of showing temporal patterns (daily, weekly, monthly) - at the expense of showing precision on quantity.

**Priestley timeline**

Good when core and duration are key elements of the story in the data.

**Circle timeline**

Good for showing discrete values of varying size across multiple categories (eg ice hazards by continent).

**Vertical timeline**

Expands time on the Y axis. Good for displaying smaller time series that work especially well when scrolling on mobile.

**Isotype/grid**

Another alternative to the circle timeline for showing (a) the extent there are big (or small) in the data.

**Streamgraph**

A type of area chart that wraps around a baseline in a geometrical look - time is more important than individual values.

## Magnitude

Show size comparisons. These can be relative (just basing data to see largest/biggest) or absolute (need to see the differences). Usually these show a 'counted' number (for example, barrels, dollars or people) rather than a calculated one or per cent.

**Example FT uses**
Commodity production, market capitalisation, economic in general

**Column**

The standard way to compare the size of things. Must always start at 0 on this axis.

**Bar**

See above. Good when the data are sort-time series and labels have long category names.

**Paired column**

As per standard column but allows for multiple series. Can become tricky to read with more than 2 series.

**Paired bar**

See above.

**Marimekko**

A good way of showing the size and proportion of data all the same time - as long as the data are not too complicated.

**Proportional symbol**

Use when there are big variations between values and/or seeing fine differences. Whereas dot is not so important.

**Isotype (pictogram)**

Brochure solution as some memorable - use only with whole numbers (do not slice off an arm to represent a decimal).

**Lollipop**

Lollipop charts draw more attention to the data value than standard bar/column - does not have to start at zero (but probably).

**Radar**

A space efficient way of showing value of multiple variables - but make sure they are organised in a way that makes sense to the reader.

**Parallel coordinates**

An alternative to radar charts - again this an progression of the variables is important (usually towards) from highlighting values.

**Bullet**

Good for showing a measurement against the context of a target or performance range.

**Grouped symbol**

An alternative to bar/column charts when being able to count dots or highlight individual extremes is useful.

## Part-to-whole

Show how a single entity can be broken down into its component elements. If the reader is interested in solely in the size of the components, consider a magnitude type chart instead.

**Example FT uses**
Fiscal budgets, company structures, national election results

**Stacked column/bar**

A simple way of showing part-to-whole relationships but can be difficult to read with more than a few components.

**Marimekko**

A good way of showing the size and proportion of data all the same time - as long as the data are not too complicated.

**Pie**

A common way of showing part-to-whole data - but be aware that it's difficult to accurately compare the size of the segments.

**Donut**

Similar to a pie chart - but the centre can be a good way of making space to include more information about the data (eg total).

**Treemap**

Use for hierarchical part-to-whole relationships can be difficult to read when there are many small segments.

**Voronoi**

A way of turning points into areas - one point within each area is closer to the centre point than any other centroid.

**Arc**

A hemicycle, often used for visualising parliamentary composition by number of seats.

**Gridplot**

Good for showing % information, they work best when used on whole numbers and work well in small multiple (easier to).

**Venn**

Generally only used for symbolic representation.

**Waterfall**

Can be used in to showing part-to-whole relationships where some of the components are negative.

## Spatial

Aside from terrain maps only used when precise locations or geographical patterns in data are more important to the reader than anything else.

**Example FT uses**
Population density, natural resource locations, natural disaster risk/impact, catchment areas, variation in election results

**Basic choropleth (rate/ratio)**

The standard approach for plotting data on a map - it should always be rates rather than totals and use a sensible base geography.

**Proportional symbol (count/magnitude)**

Use for totals rather than rates - be wary that small differences in data will be hard to see.

**Flow map**

For showing unambiguous movement across a map.

**Contour map**

For showing areas of equal value on a map. Can use elevation, colour schemes for showing +/- values.

**Equal-area cartogram**

Converting each unit on a map to a regular and equally sized shape - good for representing voting regions with equal value.

**Scaled cartogram (value)**

Distorting and shrinking a map so that each area is sized according to a particular value.

**Dot density**

Good to show the location of individual events/locations - reader sure to annotate any patterns the reader should see.

**Heat map**

Grid-based data values mapped with an internally colour scale. As cartogram map - but not snapped to an administrative unit.

## Flow

Show the relative volumes or intensity of movement between two or more states or conditions. These might be logical sequences or geographical locations.

**Example FT uses**
Movement of funds, trade, migrants, network, relationships, relationship graphs

**Sankey**

Shows changes in flows from one condition to at least one other, good for tracing the eventual outcome of a complex process.

**Waterfall**

Designed to show the sequencing of data through a flow process, typically budgets. Can include +/- components.

**Chord**

A complex but powerful diagram which can illustrate 2-way flows (and reciprocals) in a matrix.

**Network**

Used for showing the strength and inter-connectedness of relationships of varying types.

© Financial Times




What's **interesting** about the data?

What **question** is your chart answering?

Do you show **absolute** values?
**Rates** of change?
**Ratios**?

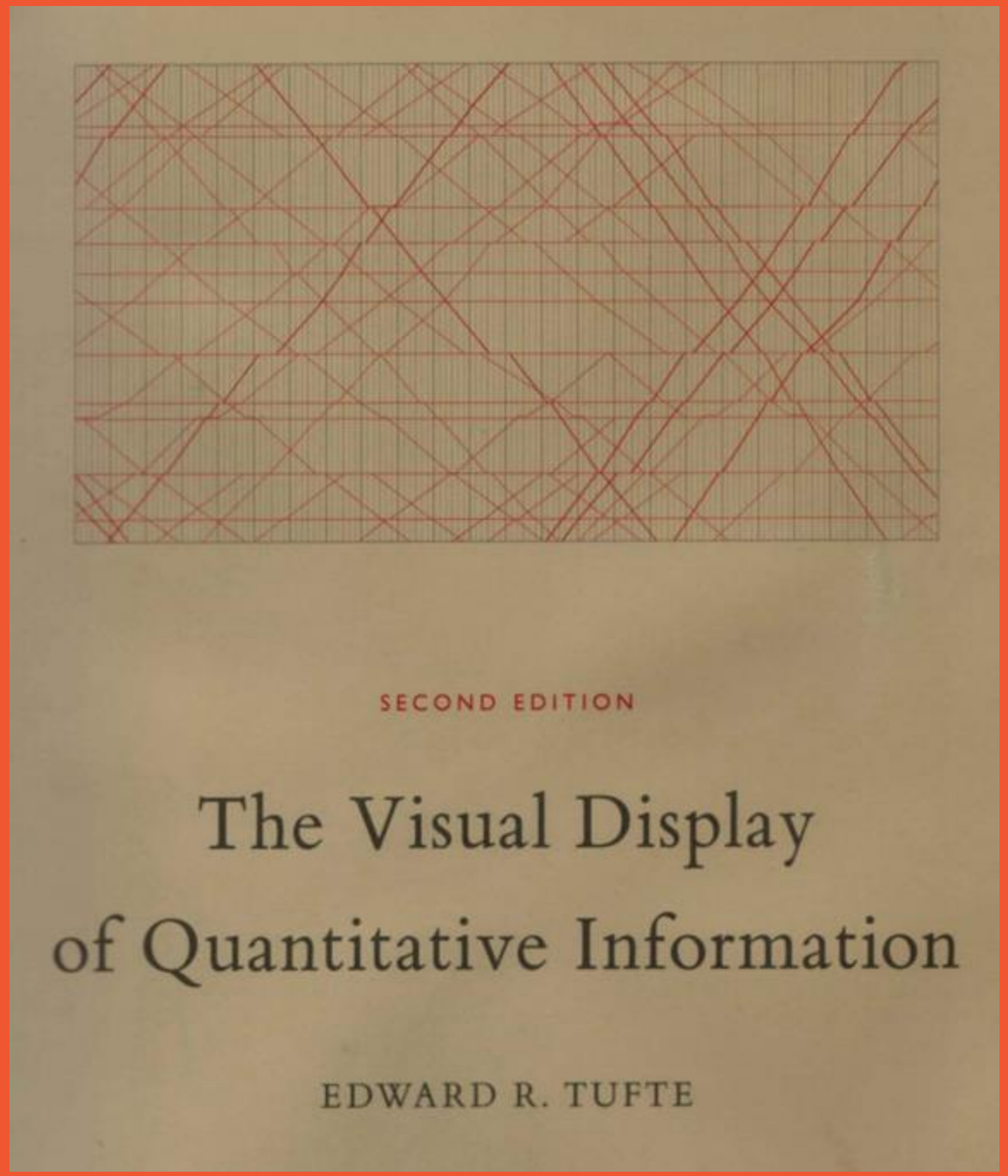
# Common Industry Tools

- Tableau
- PowerBI
- Excel
- R
- Python

Every tool has its strengths & its limitations.

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# Tufte's tips


SECOND EDITION
The Visual Display
of Quantitative Information
EDWARD R. TUFTE

*“Data graphics visually display measured quantities by means of the combined use of points, lines, a coordinate system, numbers, symbols, words, shading, and color.”*

- **Edward Tufte**

# PRINCIPLE #2:
# GRAPHICS REVEAL DATA.

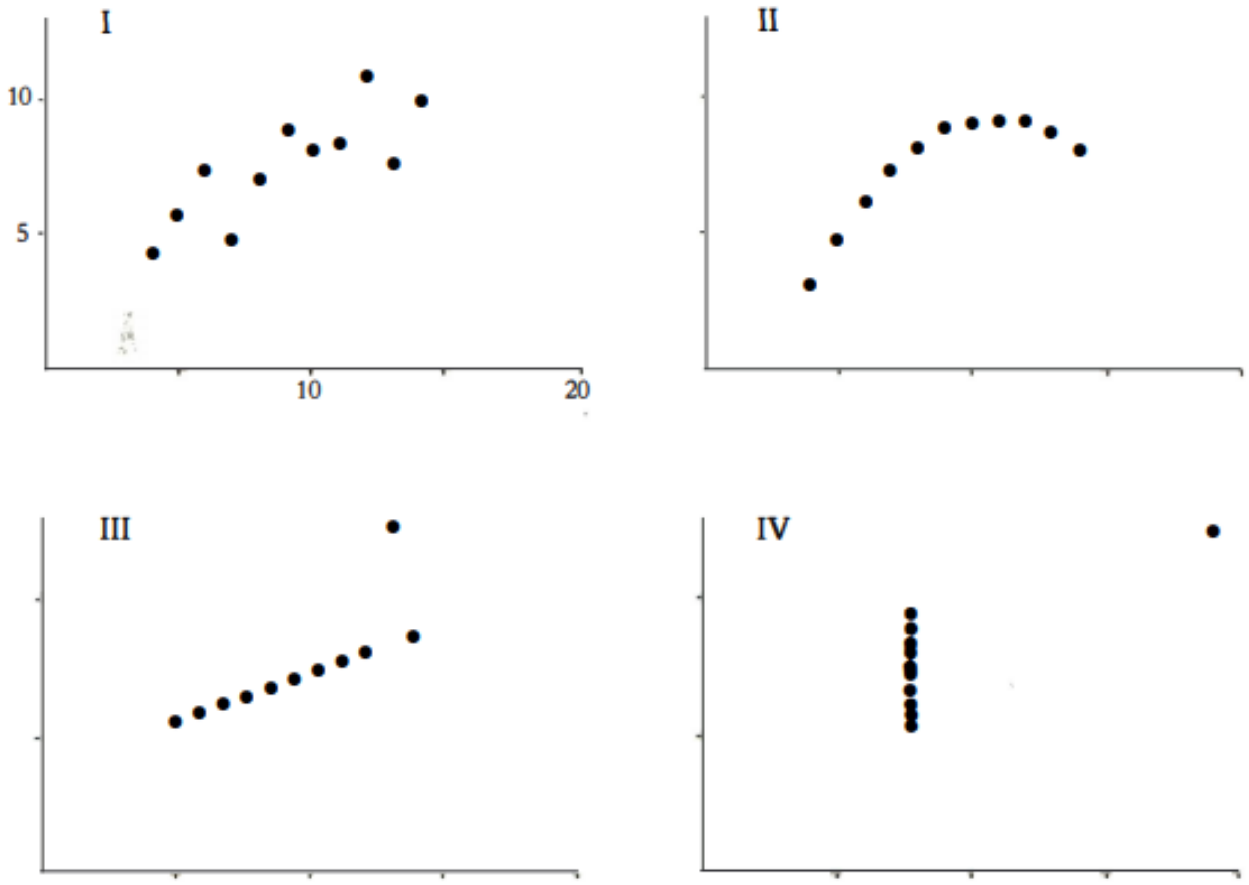
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# Anscombe's quartet

| I | | II | | III | | IV | |
|---|---|---|---|---|---|---|---|
| X | Y | X | Y | X | Y | X | Y |
| 10.0 | 8.04 | 10.0 | 9.14 | 10.0 | 7.46 | 8.0 | 6.58 |
| 8.0 | 6.95 | 8.0 | 8.14 | 8.0 | 6.77 | 8.0 | 5.76 |
| 13.0 | 7.58 | 13.0 | 8.74 | 13.0 | 12.74 | 8.0 | 7.71 |
| 9.0 | 8.81 | 9.0 | 8.77 | 9.0 | 7.11 | 8.0 | 8.84 |
| 11.0 | 8.33 | 11.0 | 9.26 | 11.0 | 7.81 | 8.0 | 8.47 |
| 14.0 | 9.96 | 14.0 | 8.10 | 14.0 | 8.84 | 8.0 | 7.04 |
| 6.0 | 7.24 | 6.0 | 6.13 | 6.0 | 6.08 | 8.0 | 5.25 |
| 4.0 | 4.26 | 4.0 | 3.10 | 4.0 | 5.39 | 19.0 | 12.50 |
| 12.0 | 10.84 | 12.0 | 9.13 | 12.0 | 8.15 | 8.0 | 5.56 |
| 7.0 | 4.82 | 7.0 | 7.26 | 7.0 | 6.42 | 8.0 | 7.91 |
| 5.0 | 5.68 | 5.0 | 4.74 | 5.0 | 5.73 | 8.0 | 6.89 |

$N = 11$
mean of X's $= 9.0$
mean of Y's $= 7.5$
equation of regression line: $Y = 3 + 0.5X$
standard error of estimate of slope $= 0.118$
$t = 4.24$
sum of squares $X - \overline{X} = 110.0$
regression sum of squares $= 27.50$
residual sum of squares of $Y = 13.75$
correlation coefficient $= .82$
$r^2 = .67$

# Anscombe’s quartet

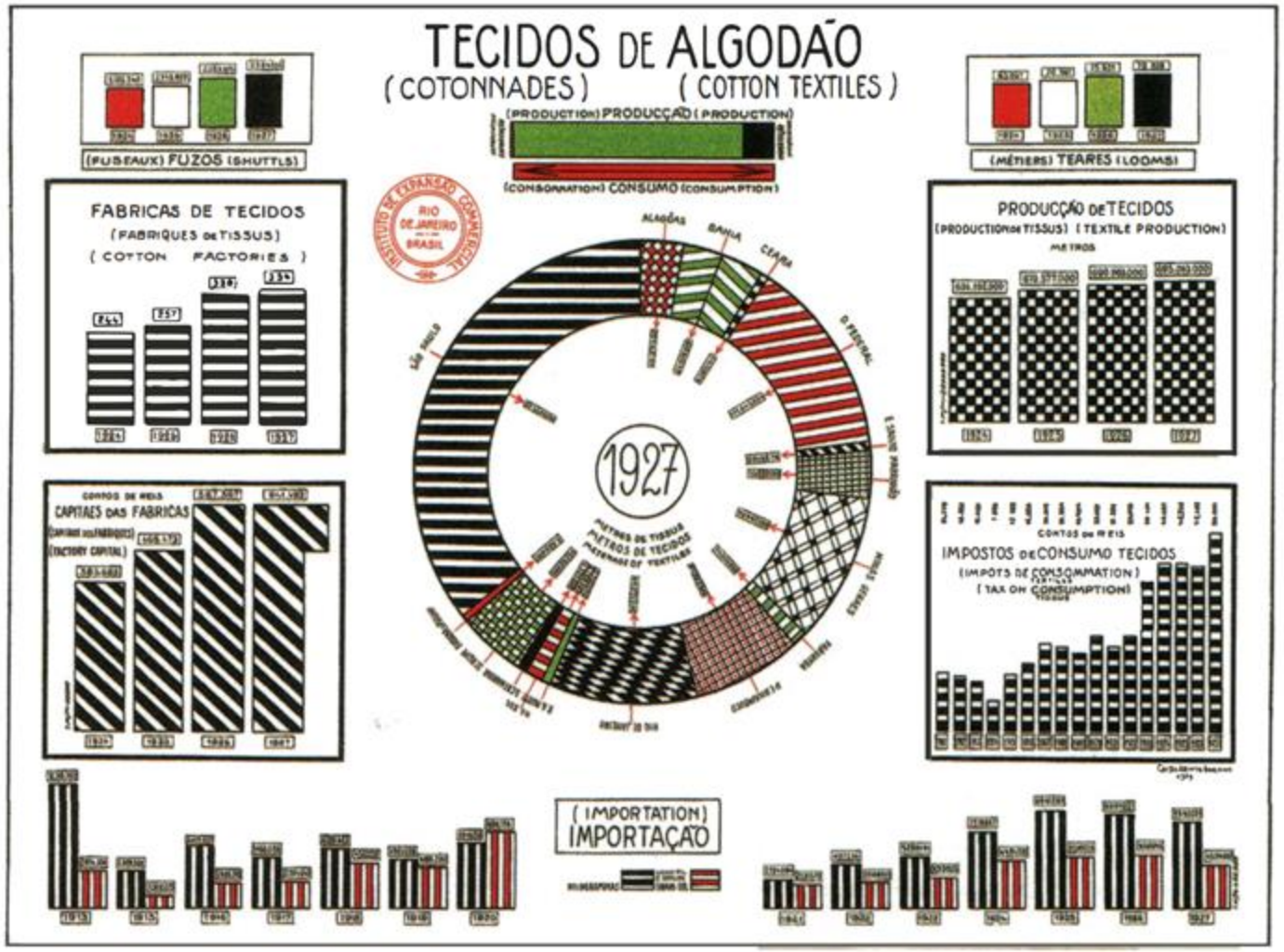


# PRINCIPLE #3:
# LESS IS MORE.

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# The Data-Ink Ratio

Avoid chart-junk.



# Remove

to improve

(the **data-ink** ratio)

Created by Darkhorse Analytics

www.darkhorseanalytics.com

# Remove to improve the pie chart edition

Created by Darkhorse Analytics

www.darkhorseanalytics.com

# PRINCIPLE #4:
# YOU DON'T ALWAYS NEED A GRAPHIC.

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“Tables are clearly the best way to show exact numerical values, although the entries can also be arranged in semi-graphical form.”

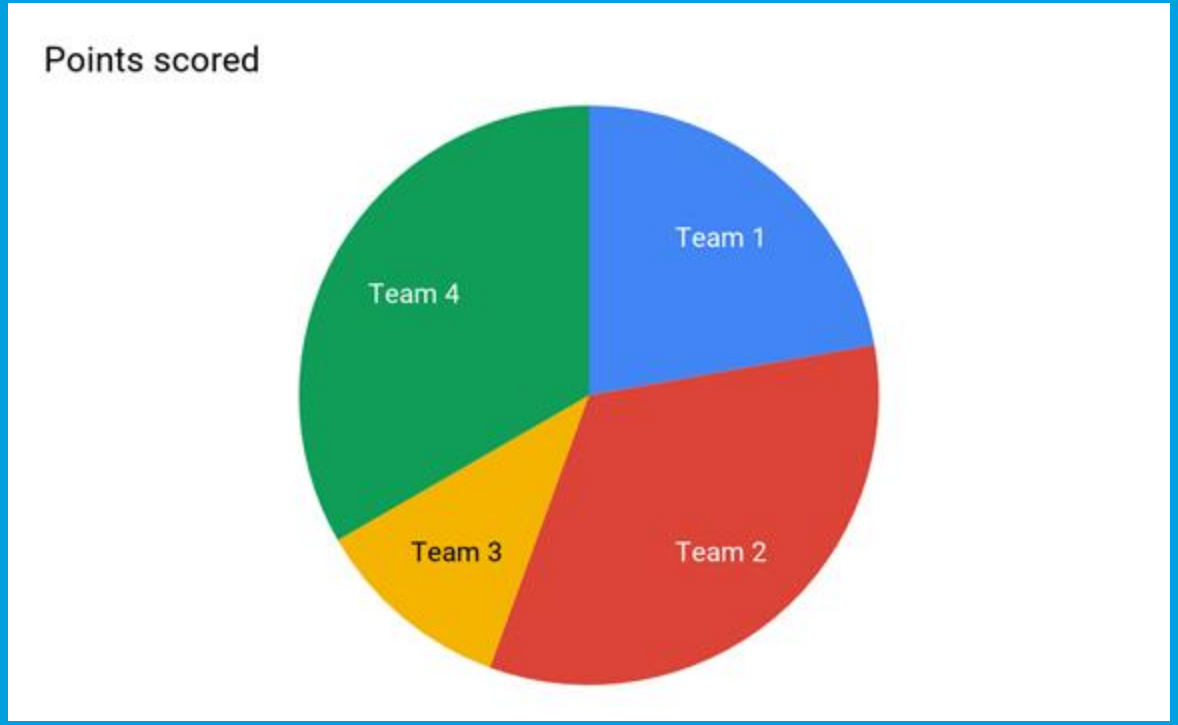
“A table is nearly always better than a dumb pie chart; the only worse design than a pie chart is several of them, for then the viewer is asked to compare quantities located in spatial disarray both within and between pies”

“Given their low data-density and failure to order numbers along a visual dimension, pie charts should never be used.”

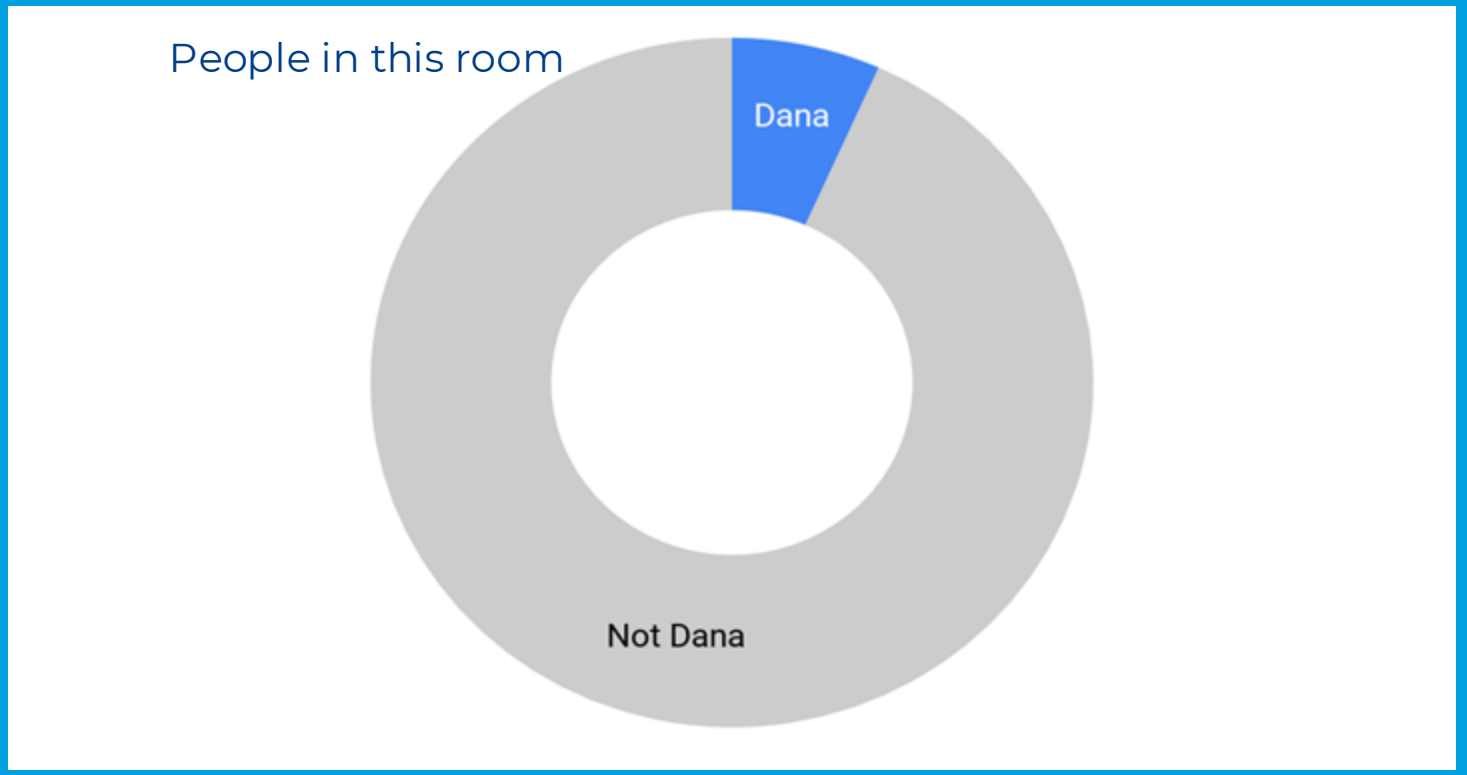
# PRINCIPLE #5:
# IF YOU THINK YOU NEED A PIE CHART, YOU PROBABLY DON'T.

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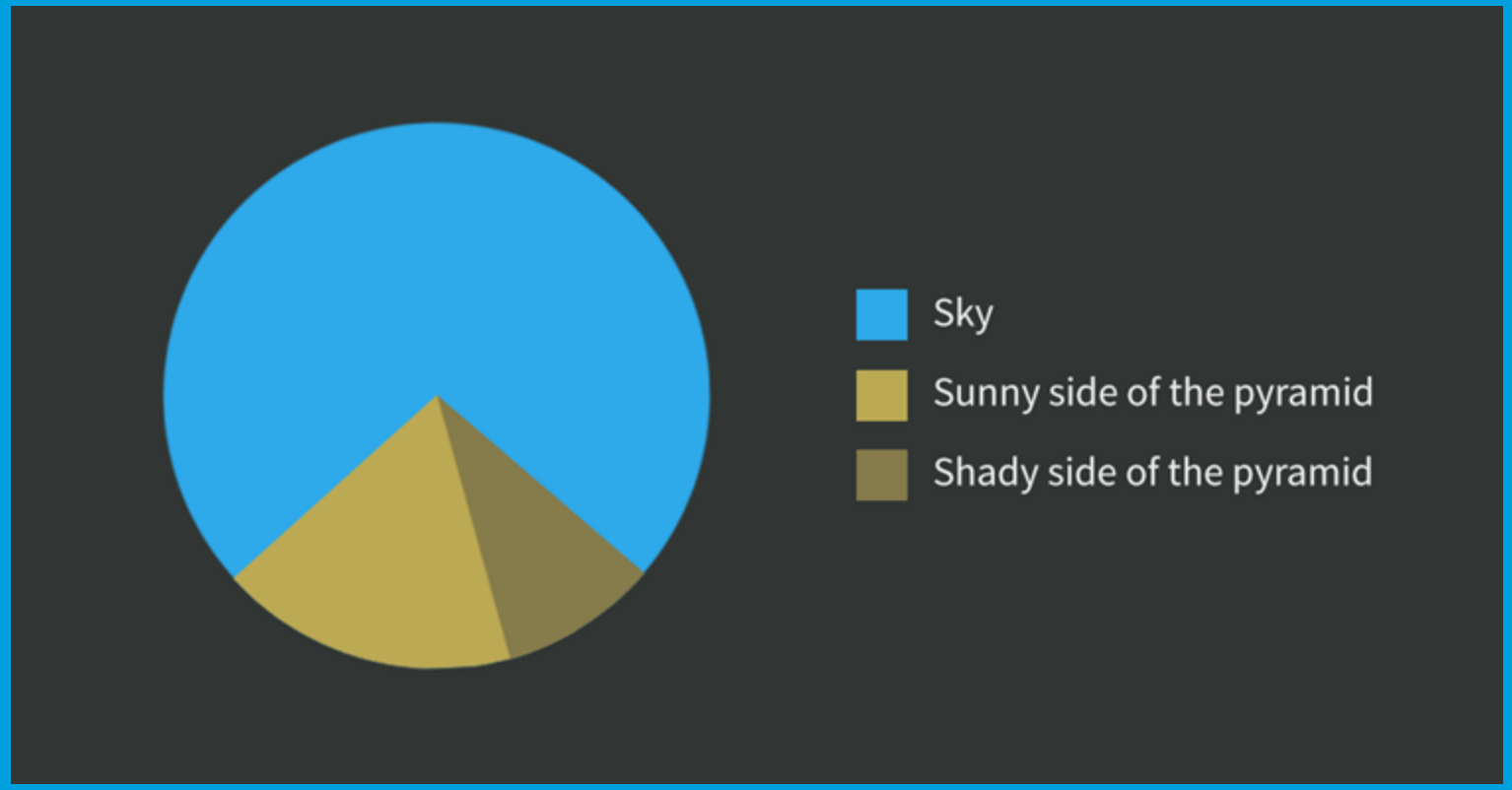
# What's the biggest slice?



# Reason to use a pie chart #1: showing a proportion



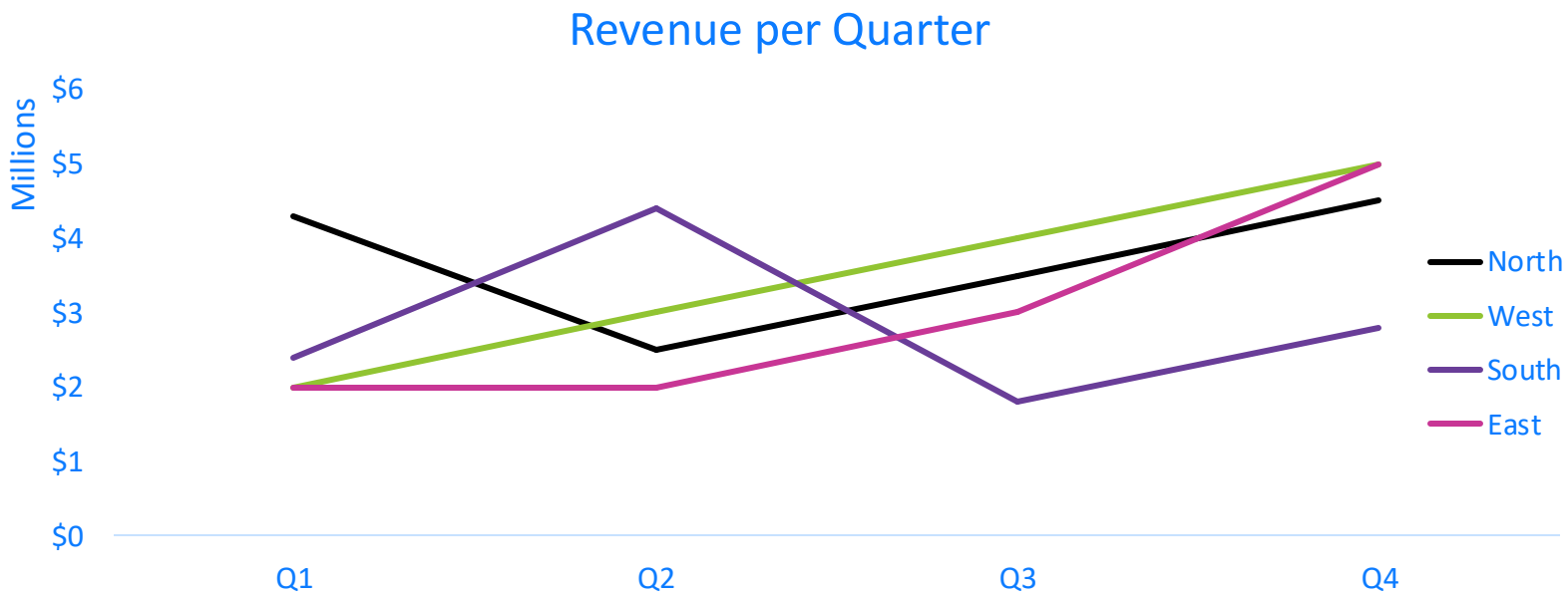
# Reason to use a pie chart #2: Making excellent jokes



# SMALL MULTIPLES

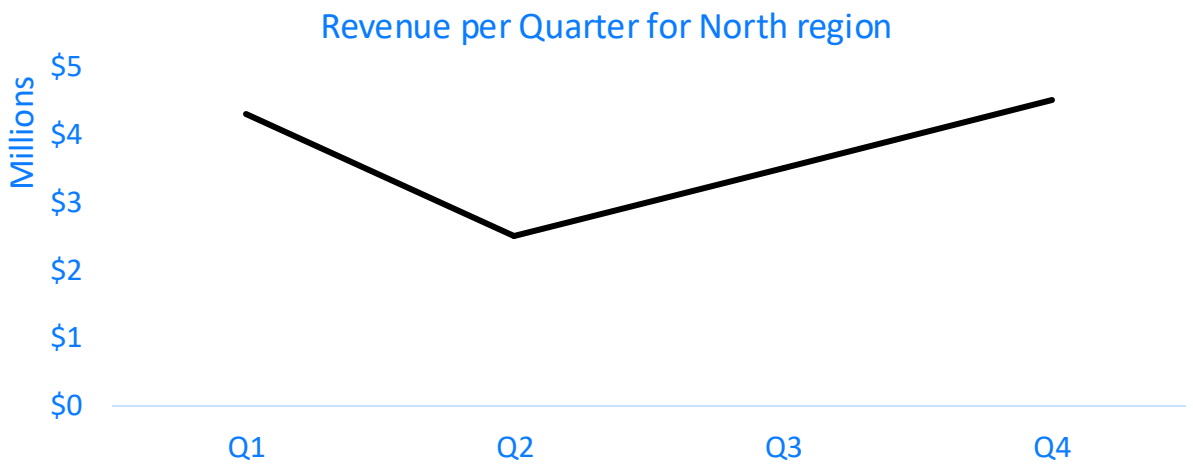
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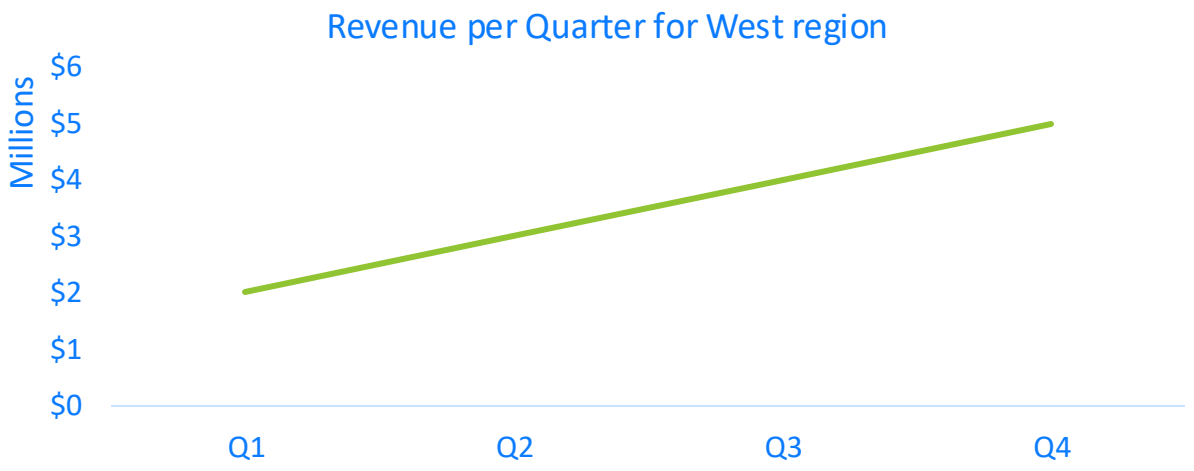
# WITHOUT SMALL MULTIPLES:

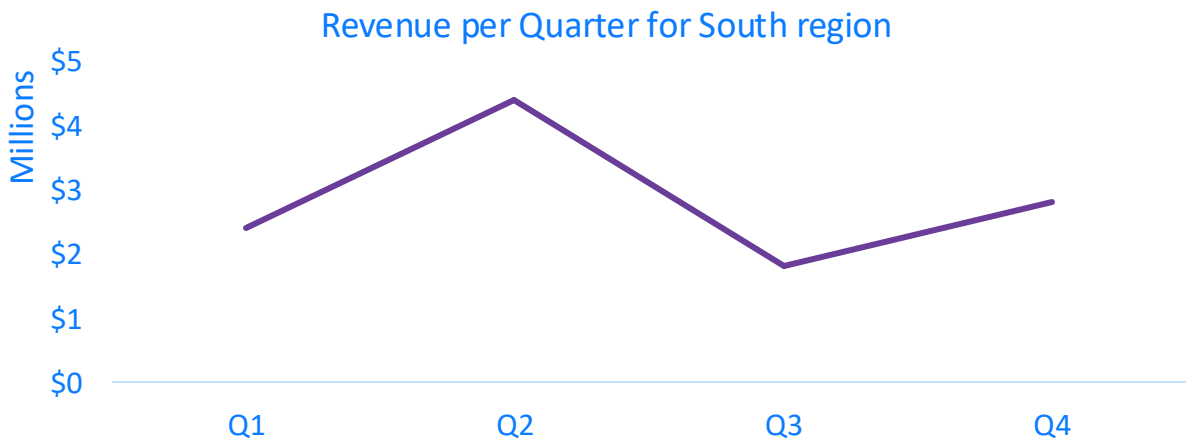


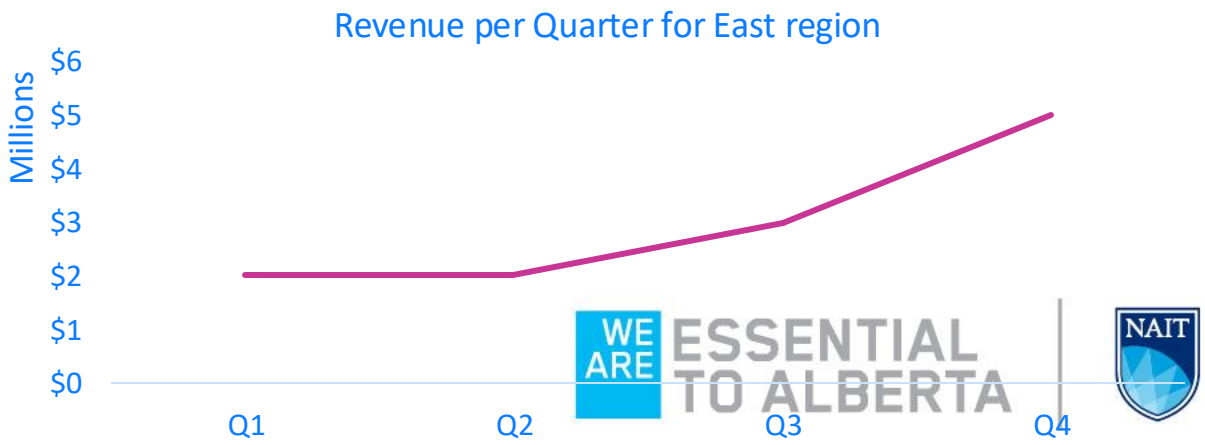

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# WITH SMALL MULTIPLES:



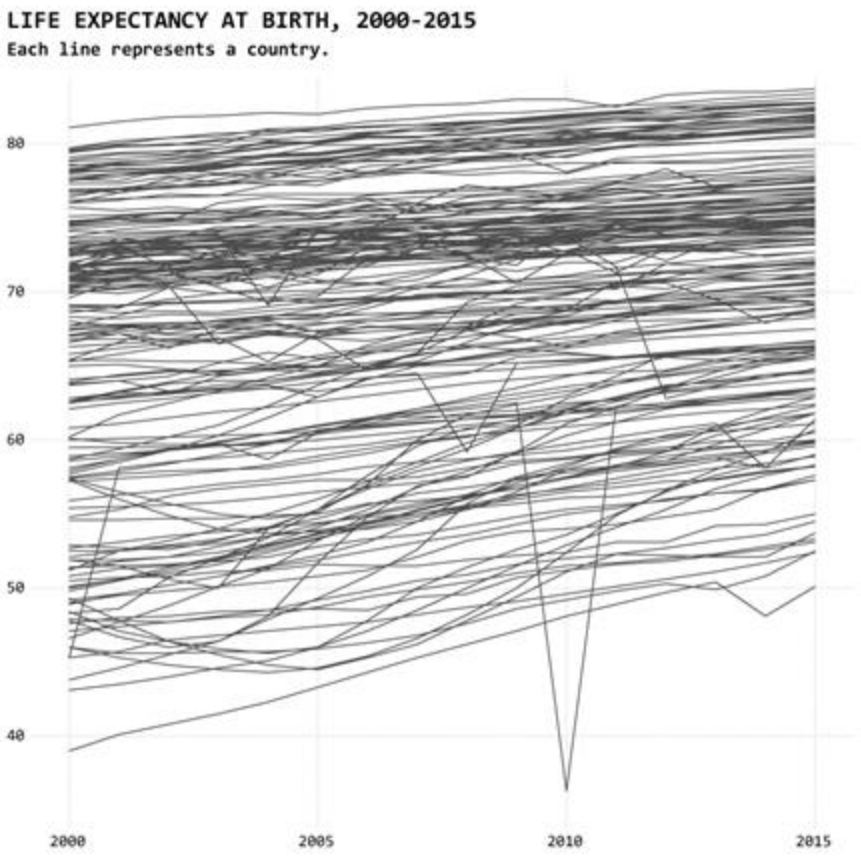



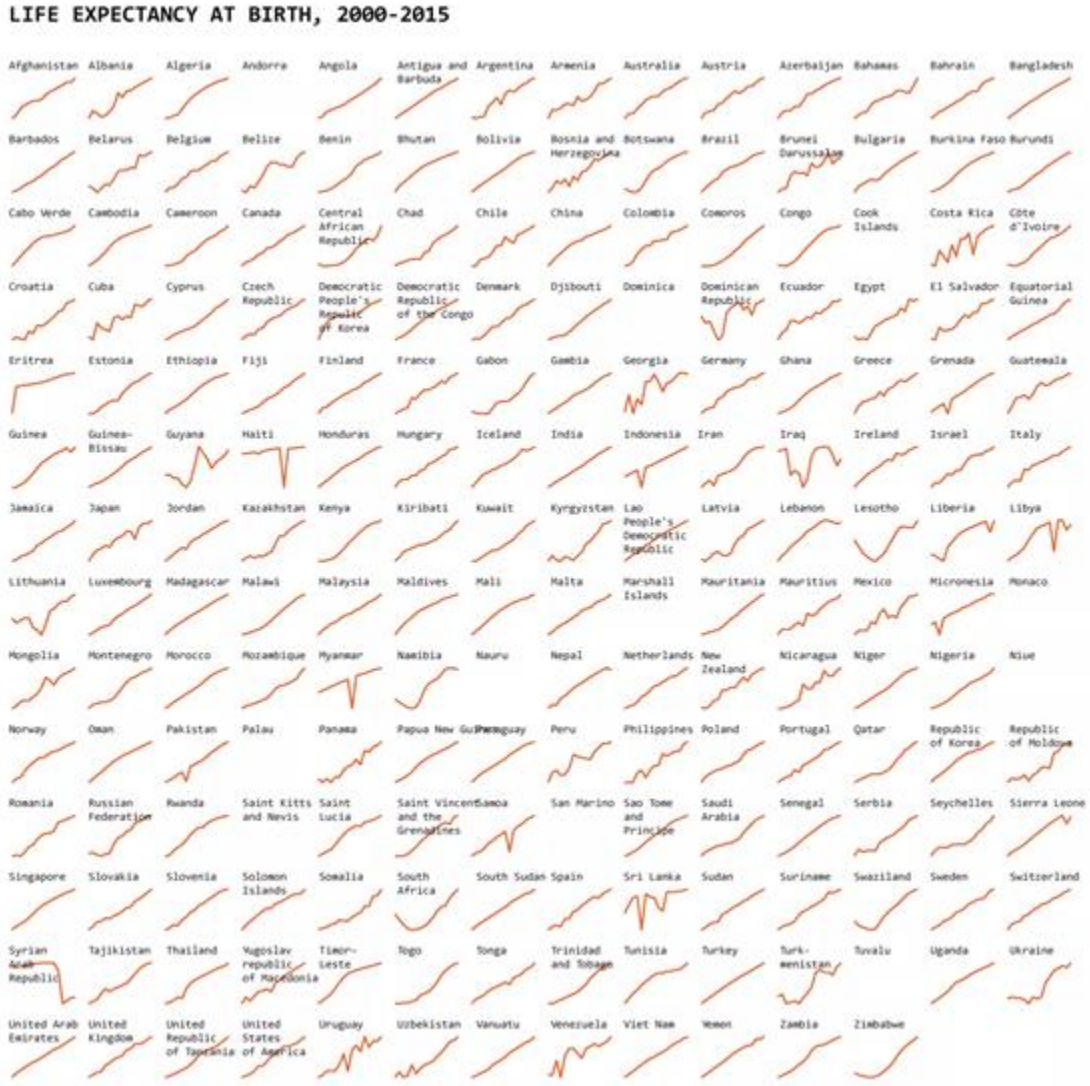




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# Sparklines


LIFE EXPECTANCY AT BIRTH, 2000-2015
Each line represents a country.
80
70
60
50
40
2000
2005
2010
2015


LIFE EXPECTANCY AT BIRTH, 2000-2015
Afghanistan
Albania
Algeria
Andorra
Angola
Antigua and Barbuda
Argentina
Armenia
Australia
Austria
Azerbaijan
Bahamas
Bahrain
Bangladesh
Barbados
Belarus
Belgium
Belize
Benin
Bhutan
Bolivia
Bosnia and Herzegovina
Botswana
Brazil
Brunei Darussalam
Bulgaria
Burkina Faso
Burundi
Cabo Verde
Cambodia
Cameroon
Canada
Central African Republic
Chad
Chile
China
Colombia
Comoros
Congo
Cook Islands
Costa Rica
Côte d'Ivoire
Croatia
Cuba
Cyprus
Czech Republic
Democratic People's Republic of Korea
Democratic Republic of the Congo
Denmark
Djibouti
Dominica
Dominican Republic
Ecuador
Egypt
El Salvador
Equatorial Guinea
Eritrea
Estonia
Ethiopia
Fiji
Finland
France
Gabon
Gambia
Georgia
Germany
Ghana
Greece
Grenada
Guatemala
Guinea
Guinea-Bissau
Guyana
Haiti
Honduras
Hungary
Iceland
India
Indonesia
Iran
Iraq
Ireland
Israel
Italy
Jamaica
Japan
Jordan
Kazakhstan
Kenya
Kiribati
Kuwait
Kyrgyzstan
Lao People's Democratic Republic
Latvia
Lebanon
Lesotho
Liberia
Libya
Lithuania
Luxembourg
Madagascar
Malawi
Malaysia
Maldives
Mali
Malta
Marshall Islands
Mauritania
Mauritius
Mexico
Micronesia
Monaco
Mongolia
Montenegro
Morocco
Mozambique
Myanmar
Namibia
Nauru
Nepal
Netherlands
New Zealand
Nicaragua
Niger
Nigeria
Niue
Norway
Oman
Pakistan
Palau
Panama
Papua New Guinea
Paraguay
Peru
Philippines
Poland
Portugal
Qatar
Republic of Korea
Republic of Moldova
Romania
Russian Federation
Rwanda
Saint Kitts and Nevis
Saint Lucia
Saint Vincent and the Grenadines
Samoa
San Marino
Sao Tome and Principe
Saudi Arabia
Senegal
Serbia
Seychelles
Sierra Leone
Singapore
Slovakia
Slovenia
Solomon Islands
Somalia
South Africa
South Sudan
Spain
Sri Lanka
Sudan
Suriname
Swaziland
Sweden
Switzerland
Syrian Arab Republic
Tajikistan
Thailand
Yugoslav Republic of Macedonia
Timor-Leste
Togo
Tonga
Trinidad and Tobago
Tunisia
Turkey
Turkmenistan
Tuvalu
Uganda
Ukraine
United Arab Emirates
United Kingdom
United Republic of Tanzania
United States of America
Uruguay
Uzbekistan
Vanuatu
Venezuela
Viet Nam
Yemen
Zambia
Zimbabwe

# Key design “rules”

| | |
|---|---|
| Simplify | Limit chart junk<br>• e.g. remove redundant labels |
| Guide | Consider the user’s path through the viz<br>• e.g. use direct labels |
| Visuals | Choose an effective visual<br>• e.g. pie charts vs line charts vs bar/column charts vs tables |
| Colour | Use colour effectively<br>• e.g. does a colour always mean the same thing? |



# Further reading….

- Data Points: Visualization That Means Something by Nathan Yau
- Functional Art: An introduction to information graphics and visualization by Alberto Cairo
- Information Dashboard Design: Displaying Data for At-a-Glance Monitoring by Stephen Few
- Now You See It: Simple Visualization Techniques for Quantitative Analysis by Stephen Few
- Show Me The Numbers: Designing Tables & Graphs to Enlighten by Stephen Few
- Storytelling with Data: A Data Visualization Guide for Business Professionals by Cole Nussbaumer Knaflic
- Truthful Art: Data, Charts, and Maps for Communication by Alberto Cairo
- The Visual Display of Quantitative Information by Edward Tufte
- The Wall Street Journal Guide to Information Graphics: The Dos and Don'ts of Presenting Data, Facts, and Figures by Dona M. Wong

# What’s next?

- Continuous and Discrete Visualization

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